Differential equations

1. Linear dif. eq. with constant coefficients
2. Matrix linear homogeneous equations.

Fundamental solution set. Fundamental matrix.

3. Wronskian, linear independence of functions
4. Fourier series.

Cheat sheet: standard size, both sides, handwritten.

Place:

160 Anthropology/Art building for people whose last name starts with A, B, C, D

155 Dwinelle, all others.

Time: 7-10 PM Wednesday, May 13

Next week: Review of linear algebra part

Monday, May 4.

Office hours: M, F 10-11:30, 709 Evans.

Problem 1. (Solve initial value problem)

$$y'' + 4y = 3\cos t + t^2, \quad y(0) = 0, \quad y'(0) = 1.$$

1. Solve homogeneous equation

$r^2 + 4 = 0 \qquad r^2 = -4 \qquad r_{1,2} = \pm 2i$

$\alpha + \beta i \qquad e^{\alpha t}\cos\beta t, \; e^{\alpha t}\sin\beta t$

$\cos 2t, \; \sin 2t$

General solution of homog. equation

$$c_1 \cos 2t + c_2 \sin 2t$$

② Find particular solution $y'' + 4y = \underbrace{3\cos t} + \underbrace{t^2}$

$y'' + 4y = 3\cos t$ $\qquad A\cos t + B\sin t$

$(A\cos t + B\sin t)'' + 4(A\cos t + B\sin t) = 3\cos t$

$-A\cos t - B\sin t + 4A\cos t + 4B\sin t = 3\cos t$

$(-A + 4A)\cos t = 3\cos t$ $\qquad A = 1$

$(-B + 4B)\sin t = 0$ $\qquad B = 0$

$\boxed{\cos t}$ $\qquad y'' + 4y = t^2$ $\qquad At^2 + Bt + C$

$(At^2 + Bt + C)'' + 4(At^2 + Bt + C) = t^2$

$2A + 4At^2 + 4Bt + 4C = t^2$ $\qquad \boxed{\frac{t^2}{4} - \frac{1}{8}}$

$(2A + 4C) = 0$ $\qquad 4At^2 = t^2$

$4B = 0$ $\qquad A = \frac{1}{4}$ $\qquad C = -\frac{1}{8}$

③ General solution $y_p = \cos t + \frac{t^2}{4} - \frac{1}{8}$

$$y = \cos t + \frac{t^2}{4} - \frac{1}{8} + c_1 \cos 2t + c_2 \sin 2t$$

④ Find coefficcents c_1, c_2 to satisfy initial conditions $y(0), y'(0) = 1$

$$y(0) = 1 - \frac{1}{8} + c_1 = 0 \qquad c_1 = \frac{-7}{8}$$

$$y'(0) = -\sin t + \frac{t}{2} + (-2c_1 \sin 2t) + 2c_2 \cos 2t \Big|_{t=0}$$

$$= 2c_2 = 1 \qquad c_2 = \frac{1}{2}$$

$$y = \cos t + \frac{t^2}{4} - \frac{1}{8} - \frac{7}{8} \cos 2t + \frac{1}{2} \sin 2t$$

Problem 2 Find a general solution of the equation $y'' + 3y' + 2y = e^{-t}$

① homogeneous equation $y'' + 3y' + 2y = 0$

$r^2 + 3r + 2 = (r+1)(r+2) = 0 \quad r_1 = -1, \; r_2 = -2$

$c_1 e^{-t} + c_2 e^{-2t}$

② Particular solution $y'' + 3y' + 2y = e^{-t}$

$$(Ate^{-t})'' + 3(Ate^{-t})' + 2Ate^{-t} =$$

$$= -2Ae^{-t} + \underbrace{Ate^{-t}} + 3Ae^{-t} \underbrace{-3Ate^{-t}} + \underbrace{2Ate^{-t}} = e^{-t}$$

$$Ae^{-t} = e^{-t} \qquad A = 1$$

$$y_p = te^{-t}$$

③ General solution: $\boxed{te^{-t} + c_1 e^{-t} + c_2 e^{-2t}}$

$$(fg)'' = \underbrace{f''g} + 2f'g' + fg''$$

Problem 3. Find a general solution and fundamental matrix of the system

$$\underline{x}'(t) = \begin{bmatrix} 3 & 5 \\ 5 & 3 \end{bmatrix} \underline{x}(t)$$

1. Diagonalize $\begin{vmatrix} 3-\lambda & 5 \\ 5 & 3-\lambda \end{vmatrix} = (\lambda-3)^2 - 5^2 =$

$= (\lambda-8)(\lambda+2) = 0 \qquad r_1 = 8, \quad r_2 = -2$

Eigen vectors $r_1 = 8 \quad \begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} \underline{u}_1 = 0 \quad \underline{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Eigen vector $r_2 = -2 \quad \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \underline{u}_2 = 0 \quad \underline{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\underline{x}_1 \ e^{8t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \ \underline{x}_2 = e^{-2t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(2) General solution

$$c_1 e^{8t}\begin{bmatrix}1\\1\end{bmatrix} + c_2 e^{-2t}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}c_1 e^{8t} + c_2 e^{-2t}\\ c_1 e^{8t} - c_2 e^{-2t}\end{bmatrix}$$

(3) Fundamental matrix

$$X(t) = \begin{bmatrix}\underline{x}_1(t) & \underline{x}_2(t)\end{bmatrix} = \begin{bmatrix}e^{8t} & e^{-2t}\\ e^{8t} & -e^{-2t}\end{bmatrix}$$

Problem 4. Solve the initial value problem

$$\underline{x}'(t) = \begin{bmatrix} -1 & 2 \\ -2 & -1 \end{bmatrix} \underline{x}(t), \qquad \underline{x}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

① Eigenvalues

$$\begin{vmatrix} -1-\lambda & 2 \\ -2 & -1-\lambda \end{vmatrix} = (\lambda+1)^2 + 4 = 0$$

$$(\lambda+1)^2 = \pm 2i \qquad r_1 = -1+2i, \quad r_2 = -1-2i$$

② complex eigenvector $-1+2i$

$$\begin{bmatrix} -2i & 2 \\ -2 & -2i \end{bmatrix} \underline{u} = 0 \quad \underline{u} = \begin{bmatrix} 1 \\ i \end{bmatrix} = \underline{a} + i\,\underline{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + i \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

③ Fundamental solution set

$$\underline{x}_1(t) = e^{-t} \cos 2t \begin{bmatrix} 1 \\ 0 \end{bmatrix} - e^{-t} \sin 2t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{x}_2(t) = e^{-t} \sin 2t \begin{bmatrix} 1 \\ 0 \end{bmatrix} + e^{-t} \cos 2t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$c_1 e^{-t} \begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} \sin 2t \\ \cos 2t \end{bmatrix} = \underline{x}(t)$$

$$\underline{x}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

4. General solution

5. Initial value

Problem 5. Determine whether functions $\{e^t, e^{-2t}, e^{3t}\}$ are linearly independent on $(-\infty, \infty)$.

$$c_1 e^t + c_2 e^{-2t} + c_3 e^{3t} = 0$$

Solution 1.

$t=0 \qquad c_1 + c_2 + c_3 = 0$

$t=1 \qquad e c_1 + e^{-2} c_2 + e^3 c_3 = 0$

$t=-1 \qquad e^{-1} c_1 + e^2 c_2 + e^{-3} c_3 = 0$

$$\begin{bmatrix} 1 & 1 & 1 \\ e & e^{-2} & e^3 \\ e^{-1} & e^2 & e^{-3} \end{bmatrix} \text{ invertible}$$

functions are linearly independent.

Solution 2 Wronskian.

$$\begin{vmatrix} e^{t} & e^{-2t} & e^{3t} \\ e^{t} & -2e^{-2t} & 3e^{3t} \\ e^{t} & 4e^{2t} & 9e^{3t} \end{vmatrix} \neq 0$$

$$(D-1)(D+2)(D-3) = 0$$

$$D = \frac{d}{dt}$$

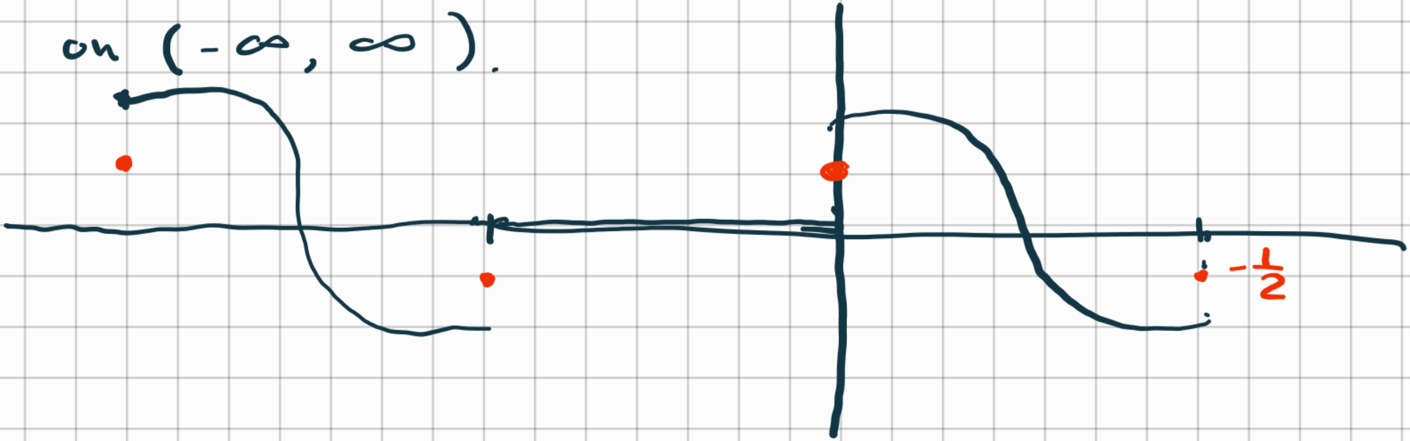
$$y''' - 2y'' - 4y' + 6y = 0$$

Solution 3

Problem 6. Compute Fourier series for

$$f(x) = \begin{cases} 0, & \pi < x < 0 \\ \cos x, & 0 < x < \pi \end{cases}$$

and draw the graph of function to which this Fourier series converges on $(-\infty, \infty)$.



Computation of Fourier coefficients

$$a_0 = \frac{1}{\pi}\int_0^{\pi} \cos x \, dx = 0$$

$$a_n = \frac{1}{\pi}\int_0^{\pi} \cos x \cos nx \, dx = 0 \quad \text{unless } n = 1$$

$$a_1 = \frac{1}{\pi}\int_0^{\pi} \cos^2 x \, dx = \frac{1}{2}$$

$$b_n = \frac{1}{\pi}\int_0^{\pi} \sin nx \cos x = \begin{cases} 0 & n \text{ is odd} \\ \dfrac{2n}{\pi(n^2-1)} & n \text{ is even} \end{cases}$$

$$f(x) = \frac{1}{2}\cos x + \frac{2}{\pi}\sum_{n=2k} \frac{n}{n^2-1} \sin(nx)$$

Formulas to use:

$$\sin\alpha \cos\beta = \frac{1}{2}\sin(\alpha+\beta) + \frac{1}{2}\sin(\alpha-\beta)$$

$$\cos\alpha \cos\beta = \frac{1}{2}\cos(\alpha+\beta) + \frac{1}{2}\cos(\alpha-\beta) \frac{1}{2}$$

Graph.